

Activity 0.1.1 Newspaper Shoes

PLTW

NEWSPAPER SHOES

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Your First Challenge

GOALS	MATERIALS	RESOURCES
• Apply a design process to creatively solve a problem. • Develop, test, and evaluate a potential solution to verify it meets project constraints and criteria. • Contribute to the efforts of a team.		
GOALS	MATERIALS	RESOURCES
• Newspaper • Rubber bands • Transparent tape • Yarn or string • Scissors		

- Stapler
- Ruler or tape measure

GOALS	MATERIALS	RESOURCES
• <u>Collaboration</u> • <u>Activity 0.1.1 Newspaper Shoes (Downloadable PDF)</u>		

In PLTW Principles of Engineering, you will complete several short, team problem-solving tasks known as Instant Challenges. In your first Instant Challenge, you experience working through a [design process](#), from defining a problem to creating and testing a prototype.

Your goal for this challenge is to create a pair of durable newspaper shoes.

Working Together

Many branches of engineering exist in today's world. As our society continues to become more technologically advanced, even more branches of engineering are created to solve specific problems and needs. Often, complex problems require engineers from varying specialties to collaborate as a team.

While working on a team can be fun, it can also be challenging. To effectively work as a team, engineers follow a common design process. That way, team members can funnel endless creativity into building working prototypes and solutions.

As you work to create a solution to today's challenge, be sure to collaborate with your teammates.



1

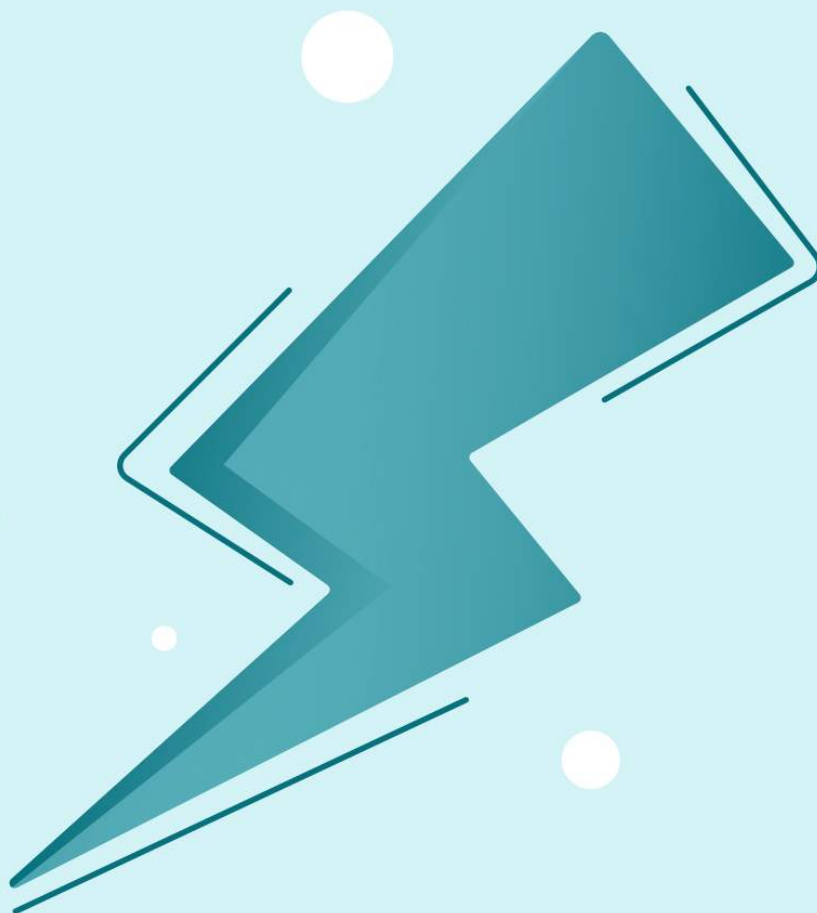
With your team, review the challenge materials and instructions, make a plan, and get started!



Engineering Notebook

Document all stages of your work in your PLTW Engineering Notebook. Describe your ideas by taking notes or creating sketches. List questions you have or problems you encountered.

NOTE: PLTW Engineering Notebook callouts serve as reminders. Even if a reminder is not present, always document work in your notebook unless instructed otherwise.



INSTANT CHALLENGE

Your Goal: Using only the materials provided, design and build a pair of newspaper shoes that properly fits one of your teammate's feet. The shoes must be durable enough to withstand a walk around the classroom. After the walk, the shoes must be taken off without significant damage.

Reflection

Collaboration is essential in the field of engineering; it improves the way a team works together and solves problems. The way a team listens, shares ideas, and shows respect to one another directly impacts the overall quality of a product. Working collaboratively also brings people closer together.

2

Review the [Collaboration](#) resource.

3

Take a moment to think about how your team worked together during today's Instant Challenge.

Turn and Talk



- Did your team use any of the steps in the collaborative problem-solving process? Which steps do you think would be useful to include in the future?
- What benefits did you gain by working as a team?
- Imagine you were to work with the same team on a future Instant Challenge. What is one change that could help the team work together more effectively?

Unit 0 Problem Introduction

Design a Shoe

Shoe For You Advertisement video (The video has no narration.) [Audio Description](#)

Throughout this course, you will wear the shoes of an engineer and use a design process to generate and test innovative solutions to authentic engineering challenges. In this unit, you focus on product design from an

engineering perspective. In the final design problem, you will design a specialized shoe for a client.

As you progress through the unit, you engage in a variety of activities and projects that explore different aspects of the product design process. Each activity helps to prepare you for solving the final problem.



Conclusion

Question 1

How did your team define success for this challenge? Were you successful according to your definition? Validate your response with evidence gathered during the challenge.

Question 2

If you had a chance to reimagine your design, how would you optimize its performance?